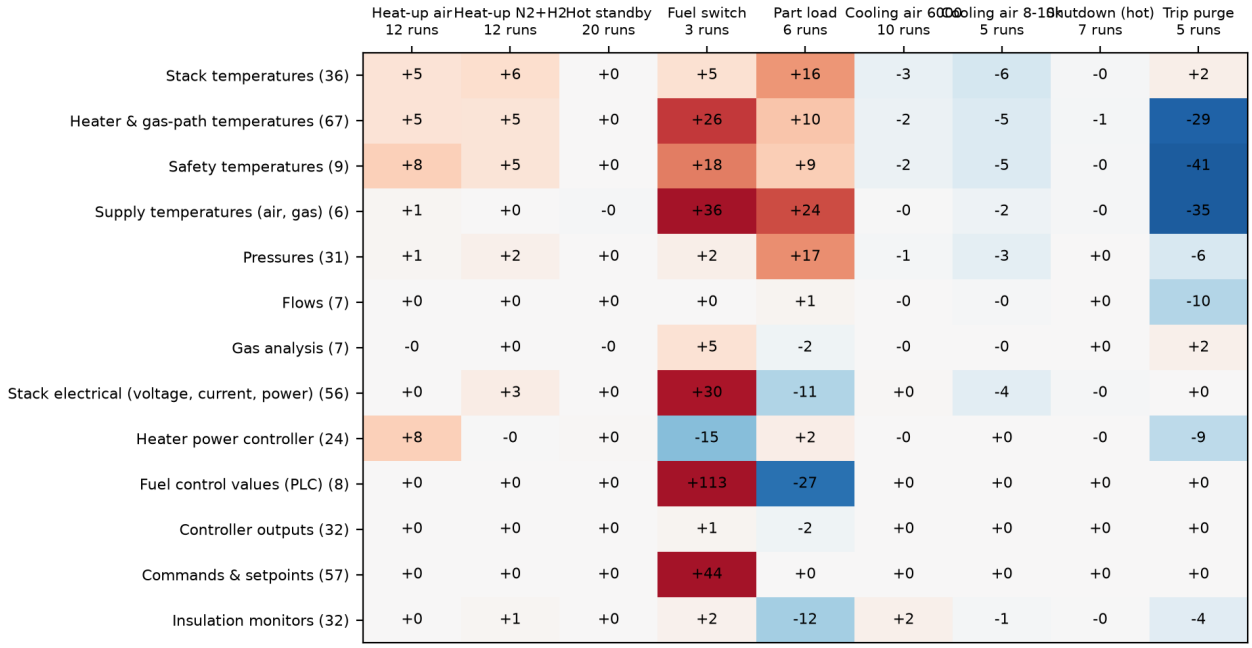
Slopes of every changing curve

SOFC plant, 39 logged days (Oct 2025 – Mar 2026). Made 2026-11-03.

Answer: the plant logs 835 columns. 622 carry live data, and **372 change with time** inside at least one stage. We measured the slope of each one in each of 9 stage groups. Temperatures and pressures follow the familiar stage pattern. The fastest changes come in fuel switch, part load and trip purge.

The slope is how fast a curve changes in one hour, in its own unit. To compare different units, the figure gives each slope as a percent of that sensor's normal full range per hour.



Typical slope in % of range per hour (median of the curves in each row). Red = rising, blue = falling.

Stage	What changes
Heat-up air / N2+H2	All temperature groups rise at 5–8 %/h. The heater power controller climbs in heat-up with air (+8.5 %/h).
Hot standby	Nothing moves (within ±0.3 %/h).
Fuel switch	Fuel-control values, commands, supply temperatures (+36 %/h) and stack electrical values (+30 %/h) jump. Only 3 runs.
Part load	String currents and power rise quickly. Stack temperatures rise at about +97 °C/h.
Cooling 6000 / 8-10k	Temperatures fall at –2 to –3 %/h with normal air and –5 to –6 %/h with strong air.
Shutdown (hot)	Slow drift, within ±1 %/h.
Trip purge	The steepest falls: heater, safety and supply temperatures drop at –28 to –41 %/h (runs of 5–20 min, extrapolated).

If you remember one thing: hot standby is flat, heat-up and cooling are slow and steady, and fuel switch, part load and trip purge move fastest.

The fastest curves in each stage

The six measured curves that change fastest relative to their range (commands, controller outputs and insulation monitors left out; at least 2 runs).

Stage	Curve	What it is	Slope per hour	% range/h
Heat-up air	TIRC_3_03	Gas temperature at the ejector outlet	+69.2 °C	+17.1
	TIRC_5_5_1	Anode off-gas temperature at TOX inlet 1	+91.4 °C	+15.8
	EIRC_2_01_ActV_rMas_Current	Heater power controller (EAPH): ActV rMas Current	+7.7 A	+15.4
	EIRC_2_01_ActV_rSlv1_Current	Heater power controller (EAPH): ActV rSlv1 Current	+7.6 A	+15.4
	EIRC_2_01_ActV_In_ThreePhasePower	Electrical power of the EAPH heater (3-phase total)	+4160 W	+14.6
	TIR_3_18	Fuel-gas temperature at the stack module inlet	+69.0 °C	+11.2
Heat-up N2+H2	pi_EIR_4_1_22	Stack voltage channel 22, string 1	-86.9 V	-9.5
	pi_EIR_4_5_22	Stack voltage channel 22, string 5	-85.5 V	-9.1
	pi_EIR_4_3_22	Stack voltage channel 22, string 3	-81.7 V	-9.0
	pi_EIR_4_2_22	Stack voltage channel 22, string 2	-79.8 V	-8.9
	pi_EIR_4_4_22	Stack voltage channel 22, string 4	-82.3 V	-8.8
	pi_EIRC_2_01_ActV_rSlv2_DeviceTemp_[°C]	Heater power controller (EAPH): ActV rSlv2 DeviceTemp [°C]	+2.8 °C	+7.5
Hot standby	pi_EIRC_2_01_ActV_rMas_Voltage_[V]	Heater power controller (EAPH): ActV rMas Voltage [V]	+6.9 V	+1.5
	TIR_1_2_2	Temperature of the POx air on its way to the reformer module	-0.30 °C	-1.4
	pi_EIRC_2_01_ActV_rMas_Power_[W]	Heater power controller (EAPH): ActV rMas Power [W]	+175 W	+1.3
	pi_EIRC_2_01_ActV_rSlv2_Power_[W]	Heater power controller (EAPH): ActV rSlv2 Power [W]	+174 W	+1.3
	pi_EIRC_2_01_ActV_rSlv1_Power_[W]	Heater power controller (EAPH): ActV rSlv1 Power [W]	+173 W	+1.3
	pi_EIRC_2_01_ActV_rMas_OutputLevel_[%]	Heater power controller (EAPH): ActV rMas OutputLevel [%]	+1.2 %	+1.3
Fuel switch	PLC_pv_rFU_Worst_2	Fuel utilisation FU Worst 2 (PLC)	+3.0 fraction	+1503
	PLC_pv_rFU_Worst_1	Fuel utilisation FU Worst 1 (PLC)	+4.0 fraction	+500
	TIR_6_01	Recycled-gas temperature after EAOH3 (ejector suction)	+920 °C	+276
	TIR_3_06	Temperature of the mixed gas at the reformer inlet	+665 °C	+258
	TIRZ_8_2_4	EAOH3 heater safety temperature (same spot as TIRC 6,1,1)	+787 °C	+252
	TIRC_6_1_1	EAOH3 heater temperature (control)	+785 °C	+226
Part load	EIRC_43_CurrentMon	String 3 load current	+15.5 A	+334
	EIRC_45_CurrentMon	String 5 load current	+14.8 A	+323
	EIRC_44_CurrentMon	String 4 load current	+11.5 A	+305
	EIRC_42_Actual_Power	Power drawn from string 2 (kW)	+6.3 kW	+292
	EIRC_45_Actual_Power	Power drawn from string 5 (kW)	+6.3 kW	+290
	EIRC_43_Actual_Power	Power drawn from string 3 (kW)	+6.0 kW	+280
Cooling air 6000	ct_FIRC_113_PV_ProcessValue	Opening / flow of the 3-way valve that splits air between the APH and the EAPH	-34.3 °C	-6.0
	EIRC_42_ActV_Min_CellVoltage	String 2 load lowest cell voltage	+0.05 V	+5.6
	pi_EIR_4_5_22	Stack voltage channel 22, string 5	+34.2 V	+3.7
	pi_EIR_4_2_22	Stack voltage channel 22, string 2	+31.8 V	+3.5
	EIRC_45_ActV_Min_CellVoltage	String 5 load lowest cell voltage	+0.04 V	+3.5
	TIRC_5_14	After-burner temperature - extra point (TIRC 5.14)	-20.1 °C	-3.3
Cooling air 8-10k	pi_EIR_4_5_22	Stack voltage channel 22, string 5	+67.4 V	+7.2
	TotalVoltage_M5	Total voltage of module 5	-34.7 V	-6.9
	TIRZ_8_1_3	Stack module safety temperature	-41.8 °C	-6.5
	TIR_4_6_33	Temperature inside the stack module - point 33 of 36	-40.1 °C	-6.5
	TIR_4_6_31	Temperature inside the stack module - point 31 of 36	-41.1 °C	-6.5
	TIR_4_6_29	Temperature inside the stack module - point 29 of 36	-40.7 °C	-6.5
Shutdown (hot)	TIR_2_1_1	Air temperature at the APH-1 inlet	+1.3 °C	+4.0
	TIR_2_1_2	Air temperature at the APH-2 inlet	+1.3 °C	+4.0
	TIR_3_05	POx-air temperature at the reformer inlet	+4.3 °C	+2.4
	TIR_2_04	Air temperature in the pipe from the 3-way valve to the APH	-0.66 °C	-2.3
	EIRC_2_01_ActV_rSlv1_MainsVoltage	Heater power controller (EAPH): ActV rSlv1 MainsVoltage	-0.34 V	-2.2
	TotalVoltage_M1	Total voltage of module 1	-9.8 V	-2.0
Trip purge	TIRZ_8_2_4	EAOH3 heater safety temperature (same spot as TIRC 6,1,1)	-988 °C	-316
	TIRC_6_1_1	EAOH3 heater temperature (control)	-1080 °C	-313
	ct_TIRC_611_PV_ProcessValue	Temperature controller 611 measured value	-1080 °C	-310

Stage	Curve	What it is	Slope per hour	% range/h
	TIR_6_01	Recycled-gas temperature after EAOH3 (ejector suction)	-934 °C	-280
	TIRC_3_09	Reformer catalyst bed temperature - point 2 of 8 inside the bed	-1250 °C	-257
	TIRC_3_10	Reformer catalyst bed temperature - point 3 of 8 inside the bed	-1250 °C	-238

If you remember one thing: in heat-up the heater side moves fastest, in part load the electrical side moves fastest, and in a trip the fuel-loop temperatures collapse.

How the slopes were measured

835 columns → 622 live → 372 that move more than 2 % of their range inside a stage (after dropping modes, status bits, alarms, counters and junk values above 10^7). For each run of a stage, a straight line is fitted through the curve. The table gives the median slope over runs. The Excel file also has the lowest and highest run.

Point	Why it matters
Commands and setpoints change in steps	Their slope is the average change per hour. They do not ramp.
Fuel switch has 3 runs; trip purge runs last 5–20 min	Treat these values as provisional. The trip values are extrapolated to one hour.
Straight line through a curved run	The value is an average over the run. The stack explainer gives rates by temperature band.
Some columns exist only on some days	A run counts only if the column has at least 80 % of its minutes.
% of range	0.5–99.5 percentile over all days. Mostly-zero signals use their full min–max range.

Files: deliverables/all-curve-slopes.xlsx (all values, own units, % of range, ranges and runs).

If you remember one thing: trust heat-up, hot standby and cooling most, and fuel switch and trip purge least.

Appendix – slope of all 372 curves (own unit per hour, median over runs)

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
Stack temperatures											
TIR_4_1	Temperature at the stack module (ICA tag TIR 4.1)	°C	+0.41	+0.13	-0.11	+3.2	+7.5	-0.03	-0.37	-0.28	+0.34
TIR_4_6_1	Temperature inside the stack module - point 1 of 36	°C	+54.5	+28.5	+0.37	+21.1	-9.8	-16.7	-30.3	-1.6	+73.0
TIR_4_6_10	Temperature inside the stack module - point 10 of 36	°C	+29.6	+39.6	+0.22	+57.2	+129	-18.2	-40.8	-1.8	+4.6
TIR_4_6_11	Temperature inside the stack module - point 11 of 36	°C	+30.0	+40.0	+0.23	+52.3	+140	-18.9	-40.5	-1.6	-23.2
TIR_4_6_12	Temperature inside the stack module - point 12 of 36	°C	+30.2	+38.8	+0.61	+25.0	+86.0	-18.2	-39.8	-1.4	+12.5
TIR_4_6_13	Temperature inside the stack module - point 13 of 36	°C	+31.2	+38.2	+0.20	+29.6	+105	-18.4	-40.6	-1.7	+10.9
TIR_4_6_14	Temperature inside the stack module - point 14 of 36	°C	+31.2	+38.4	+0.25	+47.0	+117	-18.5	-40.4	-2.0	+7.2
TIR_4_6_15	Temperature inside the stack module - point 15 of 36	°C	+30.7	+39.1	+0.21	+59.8	+120	-18.8	-40.7	-1.6	-24.9
TIR_4_6_16	Temperature inside the stack module - point 16 of 36	°C	+30.2	+39.4	+0.21	+54.9	+129	-18.2	-40.5	-1.5	-11.4
TIR_4_6_17	Temperature inside the stack module - point 17 of 36	°C	+29.2	+39.3	+0.29	+23.0	+67.3	-17.4	-40.3	-1.9	+28.3
TIR_4_6_18	Temperature inside the stack module - point 18 of 36	°C	+31.2	+38.6	+0.19	+35.0	+119	-18.5	-41.2	-1.8	+2.1
TIR_4_6_2	Temperature inside the stack module - point 2 of 36	°C	+53.7	+28.9	+0.42	+20.0	-5.2	-16.8	-30.5	-1.6	+68.0
TIR_4_6_20	Temperature inside the stack module - point 20 of 36	°C	+29.8	+39.5	+0.20	+36.7	+103	-17.4	-40.8	-1.9	+13.1
TIR_4_6_21	Temperature inside the stack module - point 21 of 36	°C	+30.6	+39.1	+0.19	+43.1	+123	-18.7	-40.7	-1.9	+13.5
TIR_4_6_22	Temperature inside the stack module - point 22 of 36	°C	+28.1	+41.3	+0.23	+41.2	+111	-16.9	-41.3	-1.6	+18.7
TIR_4_6_23	Temperature inside the stack module - point 23 of 36	°C	+26.9	+42.3	+0.26	+41.5	+113	-15.0	-41.5	-1.9	+2.4
TIR_4_6_24	Temperature inside the stack module - point 24 of 36	°C	+31.1	+38.8	+0.20	+38.3	+122	-18.4	-41.0	-1.8	+12.6
TIR_4_6_25	Temperature inside the stack module - point 25 of 36	°C	+2.0	+2.5	+0.09	+4.3	+6.9	-1.5	-3.7	-0.16	+0.60
TIR_4_6_26	Temperature inside the stack module - point 26 of 36	°C	+27.3	+42.1	+0.25	+27.4	+112	-15.8	-41.3	-1.5	+23.9
TIR_4_6_27	Temperature inside the stack module - point 27 of 36	°C	+29.1	+40.2	+0.61	+23.1	+96.2	-18.3	-40.9	-1.5	+33.6
TIR_4_6_28	Temperature inside the stack module - point 28 of 36	°C	+28.2	+40.9	+0.62	+16.2	+72.3	-16.6	-40.5	-1.3	+28.2
TIR_4_6_29	Temperature inside the stack module - point 29 of 36	°C	+30.8	+38.8	+0.19	+28.4	+77.8	-18.3	-40.7	-1.7	+8.0
TIR_4_6_3	Temperature inside the stack module - point 3 of 36	°C	+53.0	+29.5	+0.54	+17.7	+0.81	-17.0	-31.1	-1.5	+56.0
TIR_4_6_30	Temperature inside the stack module - point 30 of 36	°C	+31.4	+38.6	+0.18	+53.8	+142	-19.2	-40.7	-1.9	-7.4
TIR_4_6_31	Temperature inside the stack module - point 31 of 36	°C	+29.8	+40.0	+0.21	+35.8	+88.5	-18.4	-41.1	-1.8	+10.3
TIR_4_6_32	Temperature inside the stack module - point 32 of 36	°C	+29.2	+40.6	+0.22	+40.1	+141	-18.3	-41.2	-1.5	+18.7
TIR_4_6_33	Temperature inside the stack module - point 33 of 36	°C	+30.9	+38.4	+0.39	+9.0	+59.1	-18.9	-40.1	-1.8	+11.9
TIR_4_6_34	Temperature inside the stack module - point 34 of 36	°C	+30.8	+38.8	+0.21	+44.4	+121	-18.3	-40.6	-1.8	-8.3
TIR_4_6_35	Temperature inside the stack module - point 35 of 36	°C	+31.4	+38.7	+0.41	+26.9	+79.9	-18.7	-40.4	-1.6	+0.69
TIR_4_6_36	Temperature inside the stack module - point 36 of 36	°C	+28.7	+40.5	+0.24	+33.6	+97.1	-17.1	-40.6	-1.8	+16.2
TIR_4_6_4	Temperature inside the stack module - point 4 of 36	°C	+53.2	+29.4	+0.52	+17.8	-1.4	-16.9	-31.0	-1.5	+60.6
TIR_4_6_5	Temperature inside the stack module - point 5 of 36	°C	+55.9	+28.3	+0.54	+19.8	-8.4	-16.8	-30.1	-1.6	+67.7
TIR_4_6_6	Temperature inside the stack module - point 6 of 36	°C	+55.7	+28.4	+0.44	+19.7	-6.2	-17.0	-30.2	-1.6	+67.0
TIR_4_6_7	Temperature inside the stack module - point 7 of 36	°C	+55.5	+28.5	+0.36	+19.5	-6.9	-16.9	-30.3	-1.5	+66.5
TIR_4_6_8	Temperature inside the stack module - point 8 of 36	°C	+55.4	+28.6	+0.38	+19.5	-5.6	-16.9	-30.5	-1.5	+64.4
TIR_4_6_9	Temperature inside the stack module - point 9 of 36	°C	+30.6	+38.7	+0.20	+34.7	+112	-18.7	-40.6	-1.9	+2.6
Heater & gas-path temperatures											
TIRC_2_06	Air temperature at the EAPH outlet = air entering the stack	°C	+60.1	+26.2	+0.29	+29.2	-5.1	-15.9	-28.4	-1.6	+55.4
TIRC_3_03	Gas temperature at the ejector outlet	°C	+69.2	-1.2	-0.86	+744	+155	-0.82	-0.38	-5.7	-692
TIRC_3_07	Reformer catalyst temperature - inlet (representative)	°C	+27.0	+6.5	-0.33	+843	-59.3	-7.1	-8.1	-2.5	-852
TIRC_3_08	Reformer catalyst bed temperature - point 1 of 8 inside the bed	°C	+26.4	+7.2	-0.33	+818	-79.0	-7.2	-8.1	-1.9	-856

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
TIRC_3_09	Reformer catalyst bed temperature - point 2 of 8 inside the bed	°C	+27.1	+7.3	-0.33	+655	-83.6	-7.9	-8.9	-2.8	-1250
TIRC_3_10	Reformer catalyst bed temperature - point 3 of 8 inside the bed	°C	+27.1	+11.4	-0.23	+609	-128	-10.4	-11.4	-2.4	-1250
TIRC_3_11	Reformer catalyst bed temperature - point 4 of 8 inside the bed	°C	+27.5	+16.7	-0.20	+712	-83.3	-14.1	-17.8	-4.1	-1080
TIRC_3_12	Reformer catalyst bed temperature - point 5 of 8 inside the bed	°C	+27.9	+29.2	+0.0039	+574	-93.9	-18.6	-24.8	-3.5	-682
TIRC_3_13	Reformer catalyst bed temperature - point 6 of 8 inside the bed	°C	+29.4	+28.5	+0.14	+482	+2.4	-16.8	-32.1	-2.8	-82.9
TIRC_3_14	Reformer catalyst bed temperature - point 7 of 8 inside the bed	°C	+27.8	+36.1	+0.04	+224	+38.2	-14.4	-37.8	-5.6	-82.2
TIRC_3_15	Reformer catalyst temperature - outlet (representative)	°C	+31.3	+31.2	+0.09	+274	+51.5	-12.4	-35.1	-3.5	-10.9
TIRC_3_16	Reformer catalyst bed temperature - extra point after the catalyst	°C	+26.5	+34.7	+0.05	+200	+46.3	-11.4	-36.9	-4.3	-82.6
TIRC_3_17	Gas temperature at the reformer outlet	°C	+50.7	+33.1	+0.21	+156	+68.6	-15.3	-35.2	-4.6	-30.8
TIRC_3_1_1	EAGH heater temperature (control)	°C	+48.8	+27.4	+1.0	+112	+48.6	-15.0	-32.9	-8.4	-106
TIRC_5_07	After-burner temperature - extra point (TIRC 5.07)	°C	+29.5	+24.1	+0.27	+443	+57.4	-15.2	-27.0	-2.7	-510
TIRC_5_08	After-burner temperature - extra point (TIRC 5.08)	°C	+28.6	+30.2	+0.16	+342	+30.9	-18.1	-33.8	-4.6	-381
TIRC_5_09	After-burner temperature (representative)	°C	+28.6	+31.6	+0.50	+315	+31.3	-17.6	-35.2	-4.0	-299
TIRC_5_10	After-burner temperature - extra point (TIRC 5.10)	°C	+27.6	+27.0	+0.14	+396	+47.1	-15.7	-28.7	-2.6	-469
TIRC_5_11	After-burner temperature - extra point (TIRC 5.11)	°C	+27.6	+26.5	+0.14	+411	+56.0	-16.0	-27.8	-2.6	-496
TIRC_5_12	After-burner temperature - extra point (TIRC 5.12)	°C	+30.1	+32.5	+0.05	+283	+69.6	-18.8	-34.5	-5.2	-238
TIRC_5_13	After-burner temperature - extra point (TIRC 5.13)	°C	+29.4	+32.2	+0.05	+294	+69.0	-18.7	-34.5	-4.3	-247
TIRC_5_14	After-burner temperature - extra point (TIRC 5.14)	°C	+39.5	+31.2	+0.37	+216	+69.9	-20.1	-35.2	-4.2	-153
TIRC_5_15	After-burner temperature - extra point (TIRC 5.15)	°C	+35.8	+32.0	+0.38	+219	+65.9	-19.3	-35.8	-4.9	-158
TIRC_5_17	ERH heater temperature (after-burner housing heater)	°C	+79.7	+22.9	+6.9	+120	-48.7	-17.5	-36.4	-7.7	-323
TIRC_5_1_1	Air temperature at stack cathode outlet 1	°C	+28.7	+40.7	+0.21	+32.3	+94.0	-16.5	-40.4	-1.9	+9.0
TIRC_5_1_2	Air temperature at stack cathode outlet 2	°C	+29.7	+39.2	+0.30	+35.1	+98.3	-17.8	-40.5	-1.1	-5.9
TIRC_5_2_1	Gas temperature at stack anode outlet 1	°C	+33.2	+39.1	+1.9	+42.0	+10.9	-13.9	-35.2	-4.2	-126
TIRC_5_2_2	Gas temperature at stack anode outlet 2	°C	+22.6	+43.2	+1.5	+24.8	+41.2	-13.6	-41.3	-1.5	-19.6
TIRC_5_3_1	EAOH1 heater temperature (control)	°C	+54.2	+24.3	+1.1	+168	+13.4	-14.3	-30.8	-6.1	-331
TIRC_5_3_2	EAOH2 heater temperature (control)	°C	+55.1	+25.0	+4.5	+162	+4.6	-14.7	-32.2	-6.7	-302
TIRC_5_5_1	Anode off-gas temperature at TOX inlet 1	°C	+91.4	+14.7	-0.15	+453	+57.2	-10.1	-21.6	-10.1	-713
TIRC_5_5_2	Anode off-gas temperature at TOX inlet 2	°C	+59.5	+15.6	+0.38	+368	+57.6	-10.4	-22.7	-6.5	-663
TIRC_6_1_1	EAOH3 heater temperature (control)	°C	+28.5	-3.9	-0.13	+785	+213	-4.7	-5.1	-2.4	-1080
TIR_2_02	TIR 2 02	°C	+27.8	+33.2	+0.34	+93.7	+145	-12.4	-34.6	-4.0	-128
TIR_2_03	Air temperature after the APH (riser to the EAPH)	°C	+27.8	+33.3	+0.35	+92.0	+140	-12.4	-34.6	-4.0	-138
TIR_2_04	Air temperature in the pipe from the 3-way valve to the APH	°C	+0.42	+0.09	-0.04	+2.2	-0.26	-0.13	-0.25	-0.66	+0.10
TIR_2_05	Air temperature at the EAPH inlet	°C	+27.7	+33.3	+0.35	+90.2	+93.0	-12.3	-34.4	-1.8	-68.6
TIR_2_1_1	Air temperature at the APH-1 inlet	°C	+0.44	+0.04	-0.01	+2.1	-0.18	-0.09	-0.30	+1.3	-0.11
TIR_2_1_2	Air temperature at the APH-2 inlet	°C	+0.48	+0.04	-0.0093	+2.1	-0.23	-0.12	-0.31	+1.3	+0.43
TIR_3_01	Fuel temperature at the reformer-module inlet	°C	+0.26	+0.34	+0.27	-4.0	+1.2	-0.18	-1.0	-0.18	+26.8
TIR_3_02	Fuel temperature in the ejector main stream	°C	+0.63	+0.51	+0.48	-13.2	-2.1	-0.26	-0.67	-0.26	+16.9
TIR_3_04	POx-air temperature at the reformer-module inlet	°C	+0.27	-0.18	-0.12	+20.4	+11.1	-0.22	-0.15	+0.20	-21.9
TIR_3_05	POx-air temperature at the reformer inlet	°C	+17.0	-0.39	-0.26	-31.1	+0.24	-0.68	-0.30	+4.3	-62.4

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
TIR_3_06	Temperature of the mixed gas at the reformer inlet	°C	+26.0	-0.13	-0.76	+665	+206	-1.7	-0.97	+0.82	-473
TIR_3_18	Fuel-gas temperature at the stack module inlet	°C	+69.0	+30.8	+1.1	+98.2	+55.2	-15.3	-35.6	-7.8	-77.4
TIR_3_19_1	Fuel-gas temperature at stack manifold 1 inlet	°C	+16.2	+31.0	+1.3	+76.3	+58.7	-10.6	-33.9	-0.65	-27.6
TIR_3_19_2	Fuel-gas temperature at stack manifold 2 inlet	°C	+14.0	+31.9	+1.3	+67.1	+58.9	-10.6	-33.7	-0.66	-24.6
TIR_5_4_1	Cathode off-gas (air) temperature at TOX inlet 1	°C	+28.4	+40.3	+0.65	+40.8	+94.5	-16.6	-40.1	-5.1	-8.7
TIR_5_4_2	Cathode off-gas (air) temperature at TOX inlet 2	°C	+28.9	+39.9	+0.69	+38.3	+75.2	-16.8	-39.9	-4.9	-12.3
TIR_5_6_1	Cathode off-gas temperature in TOX bypass 1	°C	+29.6	+34.2	+0.41	+127	+109	-13.3	-35.2	-5.5	-125
TIR_5_6_2	Cathode off-gas temperature in TOX bypass 2	°C	+30.9	+33.5	+0.58	+137	+92.4	-13.7	-35.9	-5.4	-134
TIR_6_01	Recycled-gas temperature after EAOH3 (ejector suction)	°C	+31.4	-3.6	-0.96	+920	+80.4	-4.6	-4.0	-3.3	-934
TIR_7_04	Exhaust temperature at the common APH outlet	°C	+2.6	+8.1	-0.39	-27.1	+130	-2.7	-8.4	+0.19	-112
TIR_7_05	Exhaust temperature at the roof exhaust	°C	+2.6	+8.1	-0.18	-27.6	+122	-2.7	-8.2	-0.97	-111
TIR_7_1_1	Exhaust temperature at TOX outlet 1	°C	+28.8	+36.2	+0.27	+168	+94.2	-16.1	-37.6	-5.6	-138
TIR_7_1_2	Exhaust temperature at TOX outlet 2	°C	+29.4	+33.9	+0.18	+260	+64.0	-18.6	-36.3	-5.4	-219
TIR_7_2_1	Exhaust temperature after the bypass joins (mixed 1)	°C	+29.2	+34.1	+0.05	+202	+85.7	-17.9	-36.0	-6.8	-164
TIR_7_2_2	Exhaust temperature after the bypass joins (mixed 2)	°C	+29.8	+33.8	+0.14	+206	+70.9	-17.1	-36.4	-6.8	-179
TIR_7_3_1	Exhaust temperature at APH-1 outlet	°C	+1.8	+6.7	-0.49	-23.0	+110	-2.3	-7.0	+0.67	-67.4
TIR_7_3_2	Exhaust temperature at APH-2 outlet	°C	+1.8	+6.8	-0.51	-22.3	+109	-2.2	-6.8	+0.67	-57.0
ct_TIRC_206_PV_ProcessValue	Temperature controller 206 measured value	°C	+60.1	+26.2	+0.29	+29.2	-5.1	-15.9	-28.4	-1.6	+55.4
ct_TIRC_311_PV_ProcessValue	Temperature controller 311 measured value	°C	+48.8	+27.4	+1.0	+112	+48.6	-15.0	-32.9	-5.5	-106
ct_TIRC_517_PV_ProcessValue	Temperature controller 517 measured value	°C	+79.7	+22.9	+6.8	+120	-48.7	-17.5	-36.4	-7.7	-323
ct_TIRC_531_PV_ProcessValue	Temperature controller 531 measured value	°C	+54.2	+24.3	+1.1	+168	+13.4	-14.3	-30.8	-6.1	-331
ct_TIRC_532_PV_ProcessValue	Temperature controller 532 measured value	°C	+55.1	+25.0	+4.5	+162	+4.6	-14.7	-32.2	-6.7	-302
ct_TIRC_611_PV_ProcessValue	Temperature controller 611 measured value	°C	+28.5	-3.9	-0.13	+785	+213	-4.7	-5.1	-2.4	-1080
ctrl_ct_Reformer_PV_ProcessValue	Reformer temperature controller - measured value	°C	+46.3	+33.1	+0.08	+156	+68.6	-15.3	-31.5	-3.9	-30.8
Safety temperatures											
TIRZ_8_1_1	EAPH safety temperature (same spot as TIRC 2,06)	°C	+57.8	+27.9	+0.57	+32.2	-49.9	-15.6	-29.8	-2.6	+157
TIRZ_8_1_10	Exhaust outlet safety temperature	°C	+2.5	+8.1	-0.17	-28.0	+120	-2.6	-8.1	-0.94	-110
TIRZ_8_1_3	Stack module safety temperature	°C	+28.4	+40.7	+0.25	+52.6	+120	-16.7	-41.8	-1.6	-3.8
TIRZ_8_1_9	After-burner (TOX) safety temperature	°C	+28.4	+31.1	+0.20	+225	+82.5	-18.4	-35.6	-2.9	-326
TIRZ_8_2_1	EAGH heater safety temperature (same spot as TIRC 3,1,1)	°C	+51.4	+29.0	+1.00	+112	+53.8	-15.0	-33.4	-4.7	-108
TIRZ_8_2_2	EAOH1 heater safety temperature (same spot as TIRC 5,3,1)	°C	+55.8	+26.4	+1.6	+165	+9.5	-14.3	-31.3	-2.6	-312
TIRZ_8_2_3	EAOH2 heater safety temperature (same spot as TIRC 5,3,2)	°C	+58.5	+27.7	+4.9	+148	-2.3	-14.9	-32.9	-2.9	-278
TIRZ_8_2_4	EAOH3 heater safety temperature (same spot as TIRC 6,1,1)	°C	+27.7	-4.3	-0.71	+787	+224	-4.0	-4.9	-1.2	-988
TIRZ_8_2_5	ERH heater safety temperature (same spot as TIRC 5,17)	°C	+76.7	+23.0	+7.0	+122	-68.2	-17.2	-36.3	-5.5	-331
Supply temperatures (air, gas)											
TIR_1_1_1	Temperature of the air after the two streams merge	°C	+0.41	+0.13	-0.02	+2.4	-0.15	-0.01	-0.10	-0.10	-0.48
TIR_1_2_1	Temperature of the POx air in the supply module	°C	+0.20	+0.005	-0.02	+8.0	+4.1	-0.01	-0.42	-0.02	-1.7
TIR_1_2_2	Temperature of the POx air on its way to the reformer module	°C	-0.06	-0.01	-0.30	+7.7	+3.1	-0.19	-0.36	-0.24	-11.7
TIR_1_3	Temperature of the purge gas	°C	+0.13	-0.19	-0.10	+10.9	+7.2	-0.02	-0.31	+0.27	-17.9
TIR_1_4_1	Temperature of the fuel in the supply module	°C	+0.10	+0.05	+0.02	+4.5	+4.4	-0.0082	-0.35	-0.02	+2.3
TIR_1_4_2	Temperature of the fuel on its way to the reformer module	°C	-0.07	+0.0069	-0.25	+5.4	+1.5	-0.20	-0.36	-0.24	-12.5

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
Pressures											
PIRZ_8_1_2	Safety pressure monitor (fuel supply / control container - location st	bar	+0.00021	+4.4e-05	-0.00018	-0.04	-0.02	-0.00011	+7e-05	+3.9e-05	+0.0098
PIR_1_1_1	Pressure of the supply air before air flow controller 1	bar	-0.0067	+0.00077	-9.1e-05	-0.07	-0.09	-5.1e-05	+0.0032	-0.0041	-0.0047
PIR_1_1_2	Pressure of the air after the two streams merge	bar	+4.8	+3.1	+0.24	+3.8	+33.2	-1.6	-5.3	+0.0029	-32.2
PIR_1_1_3	Pressure of the supply air before air flow controller 2	bar	-0.0074	+0.00053	-0.00055	-0.06	-0.09	-8.2e-05	+0.003	-0.0061	-0.0095
PIR_1_2_1	Pressure of the supply air before the POx-air flow controller	bar	-0.0003	+0.0013	+0.00061	-0.26	-0.30	-0.002	-0.01	-0.0023	-0.10
PIR_1_2_2	Pressure of the POx air after the flow controller	bar	+0.17	+0.75	+0.15	+61.0	-41.7	-0.35	-1.5	+0.0019	+50.9
PIR_1_3_1	Pressure of the nitrogen supply	bar	+7.3e-05	-0.0041	-0.00021	+1.2	+0.12	+0.0011	+0.0081	+0.0054	-0.69
PIR_1_3_2	Pressure of the purge gas after the flow controller	bar	+0.0053	+0.04	+0.0088	-23.3	+0.41	-0.02	-0.12	-0.0014	+28.6
PIR_1_4_2	Pressure of the fuel after the flow controller	bar	+0.0003	+0.00074	+2.9e-05	+7.9	+2.7	-0.00035	-0.0013	-1.9e-05	-0.05
PIR_1_5	Pressure of the hydrogen supply	bar	-0.0036	+0.01	+0.00031	+0.95	+0.0041	+0.01	+0.03	-0.0018	+0.41
PIR_2_01	Air pressure at the EAPH inlet	bar	+4.3	+2.6	+0.14	+3.3	+1.5	-1.8	-4.9	+0.00085	-10.6
PIR_2_02	Air pressure at the stack module inlet (cathode side)	bar	+0.37	+0.60	+0.08	+0.58	+5.1	-0.28	-1.0	+0.0036	-2.2
PIR_3_01	Fuel pressure in the ejector main stream	bar	+0.0004	+0.0007	+3e-05	+7.8	+2.7	-0.00032	-0.0015	+3.3e-06	-0.04
PIR_3_02	Gas pressure at the ejector outlet	bar	+0.19	+0.71	+0.08	+11.8	+8.0	-0.27	-1.1	+7.7e-06	+0.54
PIR_3_03	Fuel-gas pressure at the stack module inlet	bar	+0.18	+0.59	+0.07	+3.2	+6.2	-0.23	-0.98	+0.0024	-0.35
PIR_5_1_1	Air pressure at stack cathode outlet 1	bar	+0.29	+0.49	+0.07	+0.60	+4.5	-0.23	-0.86	-0.00047	-1.9
PIR_5_1_2	Air pressure at stack cathode outlet 2	bar	+0.29	+0.49	+0.07	+0.76	+4.7	-0.23	-0.88	+0.00075	-1.6
PIR_5_2_1	Gas pressure at stack anode outlet 1	bar	+0.20	+0.51	+0.06	-3.3	+4.3	-0.21	-0.92	0	-1.3
PIR_5_2_2	Gas pressure at stack anode outlet 2	bar	+0.21	+0.52	+0.06	-3.4	+4.3	-0.22	-0.92	-0.0017	-1.3
PIR_5_3_1	Cathode off-gas pressure at TOX inlet 1	bar	+0.25	+0.47	+0.06	+0.59	+4.3	-0.21	-0.80	+4.4e-05	-2.6
PIR_5_3_2	Cathode off-gas pressure at TOX inlet 2	bar	+0.26	+0.47	+0.06	+0.61	+4.4	-0.23	-0.81	+0.00076	-2.6
PIR_5_4_1	Anode off-gas pressure at TOX inlet 1	bar	+0.17	+0.50	+0.06	-3.0	+4.1	-0.20	-0.86	+0.003	-1.7
PIR_5_4_2	Anode off-gas pressure at TOX inlet 2	bar	+0.17	+0.50	+0.06	-3.0	+4.1	-0.20	-0.86	+0.001	-1.6
PIR_6_01	Recycled-gas pressure before EAOH3	bar	+0.18	+0.53	+0.06	-4.2	+4.0	-0.21	-0.91	-0.00073	-1.4
PIR_7_02	Exhaust pressure at the common APH outlet	bar	+0.0067	+0.0065	+0.002	-0.03	+0.17	-0.01	-0.02	+0.00072	+0.02
PIR_7_03	Exhaust pressure at the roof exhaust	bar	+0.0028	+0.01	+0.0019	-0.03	+0.15	-0.01	-0.01	+0.0017	+0.07
PIR_7_1_1	Exhaust pressure at APH-1 inlet	bar	+0.11	+0.22	+0.03	+0.23	+2.8	-0.11	-0.35	+0.001	-2.7
PIR_7_1_2	Exhaust pressure at APH-2 inlet	bar	+0.09	+0.20	+0.03	+0.22	+2.8	0	-0.35	0	-2.7
PLC_pv_rDeltaPressEjec	Ejector pressure difference (PLC)	bar	0	+0.08	+0.0017	+12.0	+5.9	-0.02	0	+0.00082	0
PLC_pv_rPressEjector	Ejector pressure (PLC)	bar	+0.00031	0	+7.5e-06	+7.9	+4.2	-0.00013	0	0	0
PT02_Haug	PT02 Haug	bar	-3.1e-06	+0.00079	+0.04	-0.19	+0.0025	-0.00085	-0.00061	-1.2e-05	-1.1
Flows											
ct_FIRC_111_PV_ProcessValue	Air flow through flow controller 1 (stream 1)	ln/min	+2.9	+0.0013	+2.0	+0.41	+10.2	+0.0039	+0.0055	0	-535
ct_FIRC_112_PV_ProcessValue	Air flow through flow controller 2 (stream 2)	ln/min	+2.9	+0.00026	+2.0	+1.3	+8.5	+0.0035	-0.0047	0	-526
ct_FIRC_113_PV_ProcessValue	Opening / flow of the 3-way valve that splits air between the APH and	°C	+26.7	+26.7	-0.90	+90.2*	+93.0	-34.3	-34.4	+2.2	-82.7
ct_FIRC_12_PV_ProcessValue	Flow of POx air to the reformer	ln/min	0	0	0	+348	-194	0	0	0	0
ct_FIRC_13_PV_ProcessValue	Nitrogen flow	ln/min	+0.11	+1.9	+0.00035	-1250	+10.4	-0.0015	-3.1	0	+1090
ct_FIRC_14_PV_ProcessValue	Natural-gas flow (fuel)	ln/min	0	0	0	+93.2	+41.2	0	0	0	0
ct_FIRC_15_PV_ProcessValue	Hydrogen flow	ln/min	+0.0062	+0.10	+2.6e-06	-62.6	+0.53	-6e-07	-0.15	0	+54.9
Gas analysis											
pi_QIR_10_23_RH	QIR 10,23 sensor humidity (probably inside the enclosure)	% RH	-0.92	-0.08	+0.21	-3.5	-3.5	+0.04	+0.12	+0.39	-1.8

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
pi_QIR_10_23_Temp	QIR 10,23 sensor temperature (probably inside the enclosure)	°C	+0.33	+0.11	-0.15	+2.9	+4.8	-0.03	-0.18	-0.26	+0.61
pi_QIR_10_24_RH	QIR 10,24 sensor humidity (probably inside the enclosure)	% RH	-0.79	-0.28	+0.10	-2.6	-1.4	+0.20	+0.20	-0.12	+5.6
pi_QIR_10_24_Temp	QIR 10,24 sensor temperature (probably inside the enclosure)	°C	+0.08	+0.05	-0.08	+3.1	+0.84	-0.26	-0.25	-0.05	-5.1
pi_QIR_10_25_Press	pi QIR 10 25 Press		-0.08	0	-0.88	+461	-169	-0.17	-0.28	+0.50	+231
pi_QIR_10_25_RH	QIR 10,25 sensor humidity (probably inside the enclosure)	% RH	-0.92	0	-0.22	-150	+81.5	-0.09	-0.19	+1.5	-624
pi_QIR_10_25_Temp	QIR 10,25 sensor temperature (probably inside the enclosure)	°C	+0.0079	+0.40	-0.0052	+2.3	-0.66	-0.16	-0.21	+0.10	+2.8
Stack electrical (voltage, current, power)											
EIRC_41_Actual_Power	Power drawn from string 1 (kW)	kW	0	0	0	+0.45*	+6.2	0	0	0	0
EIRC_41_CurrentMon	String 1 load current	A	0	0	0	+0.92*	+15.1	0	0	+4.2e-06	0
EIRC_41_VoltageMon	String 1 load voltage	V	0	0	0	+1770*	-95.8	0	0	0	-0.12
EIRC_42_ActV_Min_CellVoltage	String 2 load lowest cell voltage	V	+0.0002	+0.04	+0.0047	+0.18*	-0.16	+0.05	-0.05	-0.01	-0.02
EIRC_42_Actual_Power	Power drawn from string 2 (kW)	kW	0	0	0	+0.34*	+6.3	0	0	0	0
EIRC_42_CurrentMon	String 2 load current	A	0	0	0	+0.70*	+15.4	0	0	0	0
EIRC_42_VoltageMon	String 2 load voltage	V	-0.0022	+0.01	-0.0021	+1790*	-96.2	-7.3e-05	+0.00093	-0.0019	-0.73
EIRC_43_ActV_Min_CellVoltage	String 3 load lowest cell voltage	V	+0.0002	+0.03	+0.0022	+0.18*	-0.12	+0.02	-0.05	-0.01	-0.02
EIRC_43_Actual_Power	Power drawn from string 3 (kW)	kW	0	0	0	+0.68*	+6.0	0	0	0	0
EIRC_43_CurrentMon	String 3 load current	A	0	-5.1e-07	0	+1.5*	+15.5	+5.8e-07	+9.2e-08	+1e-05	0
EIRC_43_VoltageMon	String 3 load voltage	V	0	0	0	+1730*	-91.0	0	0	0	0
EIRC_44_ActV_Min_CellVoltage	String 4 load lowest cell voltage	V	+0.0002	+0.03	+0.0022	+0.18*	-0.17	+0.02	-0.06	-0.0046	-0.02
EIRC_44_Actual_Power	Power drawn from string 4 (kW)	kW	0	0	0	+0.36*	+5.0	0	0	0	0
EIRC_44_CurrentMon	String 4 load current	A	0	0	0	+0.74*	+11.5	0	0	0	0
EIRC_44_VoltageMon	String 4 load voltage	V	0	0	0	+1780*	-94.5	0	0	0	-0.29
EIRC_45_ActV_Min_CellVoltage	String 5 load lowest cell voltage	V	+0.0002	+0.03	+0.0022	+0.18*	-0.12	+0.04	-0.06	-0.0043	-0.02
EIRC_45_Actual_Power	Power drawn from string 5 (kW)	kW	0	0	0	+0.35*	+6.3	0	0	0	0
EIRC_45_CurrentMon	String 5 load current	A	0	0	0	+0.71*	+14.8	0	0	0	0
EIRC_45_VoltageMon	String 5 load voltage	V	0	0	0	+1820*	-97.5	0	0	0	-0.18
String_1_ActMinVoltage	Lowest cell voltage in string 1	V	+0.53	+4.5	+0.07	+68.0	-16.9	+0.69	-5.3	-0.56	+13.9
String_2_ActMinVoltage	Lowest cell voltage in string 2	V	+0.36	+4.2	+0.09	+71.1	-16.9	+1.0	-4.4	-0.42	+6.2
String_3_ActMinVoltage	Lowest cell voltage in string 3	V	+0.88	+4.1	+0.01	+56.9	-13.1	+0.51	-5.8	-0.70	+6.7
String_4_ActMinVoltage	Lowest cell voltage in string 4	V	+0.12	+5.1	+0.03	+69.2	-20.1	+0.98	-5.1	-0.05	-3.8
String_5_ActMinVoltage	Lowest cell voltage in string 5	V	+0.99	+3.9	+0.05	+43.6	-11.7	+0.50	-4.5	-0.48	+14.8
TotalVoltage_M1	Total voltage of module 1	V	+4.7	+14.2	+0.17	+190	-55.1	+8.9	-28.3	-9.8	+8.7
TotalVoltage_M2	Total voltage of module 2	V	+5.0	+14.1	+0.18	+173	-52.5	+9.1	-28.9	-9.4	+1.5
TotalVoltage_M3	Total voltage of module 3	V	+4.7	+12.9	+0.12	+204	-53.9	+7.9	-28.9	-9.0	+4.3
TotalVoltage_M4	Total voltage of module 4	V	+5.4	+16.3	+0.21	+179	-50.6	+10.0	-29.1	-8.8	-9.3
TotalVoltage_M5	Total voltage of module 5	V	+6.6	+11.6	+0.07	+104	-46.2	+6.6	-34.7	-8.7	+0.14
pi_EIR_4_1_12	Voltage of stack 1 in string 1	V	+1.6	+3.4	+0.07	+43.4	-13.3	+0.43	-5.0	-0.82	+7.3
pi_EIR_4_1_15	Voltage of stack 2 in string 1	V	+0.48	+4.5	+0.07	+68.0	-16.9	+0.72	-5.6	-0.68	+13.9
pi_EIR_4_1_18	Voltage of stack 3 in string 1	V	+1.4	+3.8	-0.0045	+31.0	-10.8	+0.25	-5.4	-0.69	-8.5
pi_EIR_4_1_21	Stack voltage channel 21, string 1	V	+1.1	+6.2	+0.13	+47.2	-15.9	+0.50	-5.8	-0.54	-4.1
pi_EIR_4_1_22	Stack voltage channel 22, string 1	V	-0.0025	-86.9	+0.70	-99.5	-140	+26.0	+49.2	+0.0061	-20.3
pi_EIR_4_2_12	Voltage of stack 1 in string 2	V	+2.0	+3.1	+0.05	+33.3	-12.2	+0.28	-5.6	-0.86	-3.7

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
pi_EIR_4_2_15	Voltage of stack 2 in string 2	V	+1.1	+4.4	+0.04	+23.8	-12.2	+0.37	-6.3	-0.42	+4.7
pi_EIR_4_2_18	Voltage of stack 3 in string 2	V	+0.20	+4.6	+0.09	+71.2	-16.9	+1.0	-4.5	-0.97	+6.1
pi_EIR_4_2_21	Stack voltage channel 21, string 2	V	+1.4	+3.4	+0.05	+44.8	-13.3	+0.51	-5.3	-0.89	-5.6
pi_EIR_4_2_22	Stack voltage channel 22, string 2	V	-0.03	-79.8	-0.03	-110	-155	+31.8	+45.9	+0.0071	-22.0
pi_EIR_4_3_12	Voltage of stack 1 in string 3	V	+1.1	+4.1	+0.05	+54.9	-14.1	+0.41	-6.4	-0.70	+5.3
pi_EIR_4_3_15	Voltage of stack 2 in string 3	V	+1.2	+3.6	-0.02	+55.7	-10.8	+0.88	-4.0	-1.1	-16.8
pi_EIR_4_3_18	Voltage of stack 3 in string 3	V	+1.2	+3.2	+0.07	+36.0	-11.1	+0.36	-4.7	-1.1	+9.4
pi_EIR_4_3_21	Stack voltage channel 21, string 3	V	+1.5	+3.9	+0.04	+56.4	-15.3	+0.58	-6.9	-0.97	+6.5
pi_EIR_4_3_22	Stack voltage channel 22, string 3	V	+0.0069	-81.7	+1.1	-86.0	-139	+21.2	+39.4	-0.01	-31.5
pi_EIR_4_4_12	Voltage of stack 1 in string 4	V	+2.1	+4.0	+0.04	+37.0	-9.2	+0.51	-5.1	-1.1	+5.0
pi_EIR_4_4_15	Voltage of stack 2 in string 4	V	+1.7	+3.6	+0.05	+31.6	-9.2	+0.54	-4.3	-1.2	-4.5
pi_EIR_4_4_16	Voltage of package 1 of stack 3 in string 4	V	+0.00015	-2.3e-05	-7.4e-06	+0.0011	+0.00026	-0.00011	-0.0002	-1.4e-05	+0.00078
pi_EIR_4_4_17	Voltage of package 2 of stack 3 in string 4	V	+2.9e-05	+0.00013	-1.2e-06	+0.0015	+0.00021	-1.7e-05	-0.00022	0	+0.00093
pi_EIR_4_4_18	Voltage of stack 3 in string 4	V	+0.08	+5.0	+0.03	+69.2	-20.1	+0.98	-7.1	-0.05	-3.8
pi_EIR_4_4_21	Stack voltage channel 21, string 4	V	+1.4	+5.0	+0.10	+41.5	-12.1	+0.52	-5.9	-0.81	-6.0
pi_EIR_4_4_22	Stack voltage channel 22, string 4	V	+0.0098	-82.3	+1.3	-131	-205	+24.2	+59.2	+0.004	+7.6
pi_EIR_4_5_12	Voltage of stack 1 in string 5	V	+1.6	+3.1	+0.0099	+11.9	-10.6	+0.19	-5.1	-0.52	-9.2
pi_EIR_4_5_15	Voltage of stack 2 in string 5	V	+1.4	+3.4	+0.03	+26.6	-10.2	+0.28	-5.6	-0.32	-3.4
pi_EIR_4_5_18	Voltage of stack 3 in string 5	V	+1.2	+4.5	+0.04	+22.1	-10.7	+0.37	-8.1	-0.74	-2.0
pi_EIR_4_5_21	Stack voltage channel 21, string 5	V	+1.4	+3.9	+0.05	+43.6	-11.7	+0.50	-5.4	-0.42	+14.8
pi_EIR_4_5_22	Stack voltage channel 22, string 5	V	-0.00031	-85.5	-0.45	-88.4	-122	+34.2	+67.4	+0.0032	-25.6
Heater power controller											
EIRC_2_01_ActV_In_ThreePhasePower	Electrical power of the EAPH heater (3-phase total)	W	+4160	-842	+86.2	-12800	+10600	-1.2	+0.006	+0.24	-621
EIRC_2_01_ActV_rMas_Current	Heater power controller (EAPH): ActV rMas Current	A	+7.7	-0.78	+0.21	-22.5	+12.0	-0.00067	+1.6e-05	+0.00028	-1.1
EIRC_2_01_ActV_rMas_MainsVoltage	Heater power controller (EAPH): ActV rMas MainsVoltage	V	-0.72	+0.20	-0.19	+3.6	+0.63	+0.22	+0.17	-0.34	-0.59
EIRC_2_01_ActV_rSlv1_Current	Heater power controller (EAPH): ActV rSlv1 Current	A	+7.6	-0.79	+0.20	-22.6	+11.6	-5.6e-05	-4.3e-06	-2.1e-05	-1.1
EIRC_2_01_ActV_rSlv1_MainsVoltage	Heater power controller (EAPH): ActV rSlv1 MainsVoltage	V	-0.56	+0.20	-0.12	+3.4	+0.10	+0.23	+0.07	-0.34	-1.4
EIRC_2_01_ActV_rSlv2_Current	Heater power controller (EAPH): ActV rSlv2 Current	A	+7.7	-0.78	+0.21	-22.6	+12.0	-1.4e-06	+0.0098	+0.001	-1.7
pi_EIRC_2_01_ActV_rMas_ActValue_[%]	Heater power controller (EAPH): ActV rMas ActValue [%]	%	+10.0*	-2.1	+0.21	-47.5*	+31.0	-64.7*	-0.0001	-1.8*	+11.7
pi_EIRC_2_01_ActV_rMas_CurrInput_[mA]	Heater power controller (EAPH): ActV rMas CurrInput [mA]	mA	-0.0011*	-0.0021	+0.00038	+0.0015*	+0.00047	+0.0017*	+0.00025	+4.1e-05*	+0.0022
pi_EIRC_2_01_ActV_rMas_DeviceTemp_[°C]	Heater power controller (EAPH): ActV rMas DeviceTemp [°C]	°C	+3.7*	+2.8	-0.31	-1.6*	-1.7	-3.0*	-0.41	-0.36*	-9.9
pi_EIRC_2_01_ActV_rMas_OutputLevel_[%]	Heater power controller (EAPH): ActV rMas OutputLevel [%]	%	+9.5*	-2.1	+1.2	-43.3*	+24.2	0*	0	0*	-19.2
pi_EIRC_2_01_ActV_rMas_Power_[W]	Heater power controller (EAPH): ActV rMas Power [W]	W	+1420*	-307	+175	-6490*	+3580	-0.04*	+4.2e-06	-0.01*	-2970
pi_EIRC_2_01_ActV_rMas_VoltInput_[V]	Heater power controller (EAPH): ActV rMas VoltInput [V]	V	-0.00035*	-0.00068	+0.00017	+0.00089*	+0.00014	+0.00061*	+7.8e-05	+1.3e-05*	+0.00085
pi_EIRC_2_01_ActV_rMas_Voltage_[V]	Heater power controller (EAPH): ActV rMas Voltage [V]	V	+45.6*	-4.3	+6.9	-159*	+97.2	-225*	-0.04	-3.1*	-115
pi_EIRC_2_01_ActV_rSlv1_DeviceTemp_[°C]	Heater power controller (EAPH): ActV rSlv1 DeviceTemp [°C]	°C	+4.4*	+2.9	-0.33	-1.1*	-1.8	-3.0*	-0.44	-0.34*	-12.2
pi_EIRC_2_01_ActV_rSlv1_Power_[W]	Heater power controller (EAPH): ActV rSlv1 Power [W]	W	+1410*	-301	+173	-6430*	+3500	-0.27*	+6.8e-05	-0.13*	-2940
pi_EIRC_2_01_ActV_rSlv1_Resistance_[Ohm]	Heater power controller (EAPH): ActV rSlv1 Resistance [Ohm]	Ω	-73900000000000002146139619589095424*	+0.0085	-0.0033	-0.06*	+0.01	–	–	–	–
pi_EIRC_2_01_ActV_rSlv2_DeviceTemp_[°C]	Heater power controller (EAPH): ActV rSlv2 DeviceTemp [°C]	°C	+4.2*	+2.8	-0.12	+0.95*	-2.1	-2.6*	-0.44	-0.34*	-11.2

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
pi_EIRC_2_01_ActV_rSlv2_Power_[W]	Heater power controller (EAPH): ActV rSlv2 Power [W]	W	+1420*	-304	+174	-6570*	+3580	-4.9*	+0.0099	-2.6*	-2930
pi_EIRC_2_01_ActV_rSlv2_Resistance_[Ohm]	Heater power controller (EAPH): ActV rSlv2 Resistance [Ohm]	Ω	-705999999999999971952406618313850888*	+0.0072	-906000000000000012948920523030528	-0.04*	-0.03	–	–	–	–
pi_IIR_3_1_1_ActVal	EAGH heater current	A	+0.13	+0.0014	+0.04	+0.08	-5.4	+9.3e-06	0	+1.5e-05	-2.4
pi_IIR_5_1_1_ActVal	EAOH1 heater current	A	+0.12	-0.00065	+0.01	+0.0084	-1.1	+1.2e-05	-0.00019	+1.2e-05	-0.13
pi_IIR_5_1_2_ActVal	EAOH2 heater current	A	+0.14	+0.0011	+0.01	+0.01	-1.3	+8e-06	+1.7e-05	+2e-05	-0.53
pi_IIR_5_1_3_ActVal	pi IIR 5 1 3 ActVal	A	+0.14	+0.0069	+0.02	-0.13	-0.29	+9.1e-06	0	+2.3e-05	-0.51
pi_IIR_6_1_1_ActVal	pi IIR 6 1 1 ActVal	A	+0.42	0	+0.15	-4.0	-8.7	+4.9e-05	0	+8.6e-06	-0.89
Fuel control values (PLC)											
PLC_pv_rActAfsr	Actual fuel supply ratio (PLC)		0	0	0	+0.41	-2.2	0	0	0	0
PLC_pv_rActPoxValue	POx air value (PLC)		0	0	0	+348	-166	0	0	0	0
PLC_pv_rFU_Worst_1	Fuel utilisation FU Worst 1 (PLC)	fraction	0	0	0	+4.0	+42.9	0	0	0	0
PLC_pv_rFU_Worst_2	Fuel utilisation FU Worst 2 (PLC)	fraction	0	0	0	+3.0	+43.1	0	0	0	0
PLC_pv_rFU_Worst_3	Fuel utilisation FU Worst 3 (PLC)	fraction	0	0	0	+4.0	+42.7	0	0	0	0
PLC_pv_rFU_Worst_4	Fuel utilisation FU Worst 4 (PLC)	fraction	0	0	0	+3.2	+37.2	0	0	0	0
PLC_pv_rFU_Worst_5	Fuel utilisation FU Worst 5 (PLC)	fraction	0	0	0	+3.1	+40.0	0	0	0	0
PLC_pv_rRejectionRatio	Ejector rejection ratio (PLC)		0	0	0	+22.6	0	0	0	0	0
Controller outputs											
EIRC_41_OUT_Analog	Controller output of EIRC 41	%	0	0	0	+0.80*	+16.5	0	0	0	0
EIRC_41_OUT_Inhibit	Controller output of EIRC 41	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_42_OUT_Analog	Controller output of EIRC 42	%	0	0	0	+0.84*	+16.8	0	0	0	0
EIRC_42_OUT_Inhibit	Controller output of EIRC 42	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_42_OUT_StackCon_OnOff)	Device status of the load on string 2	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_43_OUT_Analog	Controller output of EIRC 43	%	0	0	0	+1.5*	+16.8	0	0	0	0
EIRC_43_OUT_Inhibit	Controller output of EIRC 43	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_43_OUT_StackCon_OnOff)	Device status of the load on string 3	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_44_OUT_Analog	Controller output of EIRC 44	%	0	0	0	+0.82*	+12.5	0	0	0	0
EIRC_44_OUT_Inhibit	Controller output of EIRC 44	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_44_OUT_StackCon_OnOff)	Device status of the load on string 4	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_45_OUT_Analog	Controller output of EIRC 45	%	0	0	0	+0.83*	+16.3	0	0	0	0
EIRC_45_OUT_Inhibit	Controller output of EIRC 45	%	0	0	0	+3.7*	-0.12	0	0	0	0
EIRC_45_OUT_StackCon_OnOff)	Device status of the load on string 5	%	0	0	0	+3.7*	-0.12	0	0	0	0
ct_FIRC_111_OUT_Analog	Controller output of FIRC 111	%	+2.8	0	+2.0	0	+2.5	0	0	0	-442
ct_FIRC_112_OUT_Analog	Controller output of FIRC 112	%	+2.8	0	+2.0	0	+2.8	0	0	0	-442
ct_FIRC_113_OUT_OutputValue	Controller output of FIRC 113	%	0	0	0	0*	+62.9	+0.09	+1.7	0	-71.8
ct_FIRC_113_SO_Analog	ct FIRC 113 SO Analog	%	0	0	0	0*	0	+0.09	0	-0.0029	-41.0
ct_FIRC_12_OUT_Analog	Controller output of FIRC 12	%	0	0	0	+348	-194	0	0	0	0
ct_FIRC_13_DO_SlewRateAbs	ct FIRC 13 DO SlewRateAbs	ln/min	+0.01	0	0	0	0	0	0	0	+0.17
ct_FIRC_13_OUT_Analog	Controller output of FIRC 13	%	+0.14	+1.9	0	-1250	+11.2	0	-3.1	0	+1050
ct_FIRC_14_OUT_Analog	Controller output of FIRC 14	%	0	0	0	+86.3	+41.3	0	0	0	0
ct_FIRC_14_SO_Analog	ct FIRC 14 SO Analog	%	0	0	0	+79.5	+52.4	0	0	-0.0058	-0.87
ct_FIRC_15_DO_SlewRateAbs	ct FIRC 15 DO SlewRateAbs	ln/min	+0.0006	0	0	0	0	0	0	0	+0.0083
ct_FIRC_15_OUT_Analog	Controller output of FIRC 15	%	+0.0068	+0.10	0	-62.6	+0.56	0	-0.15	0	+53.7

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
ct_TIRC_206_OUT_OutputValue	Temperature controller 206 measured value	%	+9.9	-1.9	+0.24	-30.3	+25.4	0	0	0	-1.4
ct_TIRC_311_OUT_OutputValue	Temperature controller 311 measured value	%	+3.1	0	+0.19	0	-79.3	0	0	0	-5.0
ct_TIRC_311_SO_OutValue	Temperature controller 311 measured value	°C	+0.27	0	0	0	-15.7	0	0	-0.009	0
ct_TIRC_517_OUT_OutputValue	Temperature controller 517 measured value	%	+3.6	0	+1.4	-37.9	-58.9	0	0	0	-3.3
ct_TIRC_531_OUT_OutputValue	Temperature controller 531 measured value	%	+3.1	0	+0.02	0	-18.3	0	0	0	-5.0
ct_TIRC_532_OUT_OutputValue	Temperature controller 532 measured value	%	+3.1	0	+0.02	0	-18.4	0	0	0	-5.0
ct_TIRC_611_OUT_OutputValue	Temperature controller 611 measured value	%	+3.1	0	+0.19	0	-3.2	0	0	0	-5.0
Commands & setpoints											
EIRC_2_01_SetV_bExtInhibitInp	Heater power controller (EAPH): SetV bExtInhibitInp		-0.0078	0	0	0	+0.06	0	0	0	+1.1
EIRC_2_01_SetV_rSetpointValue	Heater power controller (EAPH): SetV rSetpointValue		+9.9	-1.9	+0.24	-30.3	+25.4	0	0	0	-1.4
EIRC_41_SP_SlewRateAbs	EIRC 41 SP SlewRateAbs		0	0	0	+0.06*	-0.001	0	0	0	-0.00042
EIRC_41_Setp_Min_CellVoltage	EIRC 41 Setp Min CellVoltage	V	0	0	0	+0.46*	-0.15	0	0	0	0
EIRC_42_SP_Analog	EIRC 42 SP Analog		-0.0084	0	0	+8.8*	+14.3	0	0	0	-2.7
EIRC_42_SP_SlewRateAbs	EIRC 42 SP SlewRateAbs		0	0	0	+0.06*	-0.001	0	0	0	-0.00042
EIRC_42_Setp_Min_CellVoltage	EIRC 42 Setp Min CellVoltage	V	0	0	0	+0.46*	-0.15	0	0	0	0
EIRC_43_SP_Analog	EIRC 43 SP Analog		-0.0084	0	0	-2.2*	+14.3	0	0	0	-2.7
EIRC_43_Setp_Min_CellVoltage	EIRC 43 Setp Min CellVoltage	V	0	0	0	+0.46*	-0.15	0	0	0	0
EIRC_44_SP_Analog	EIRC 44 SP Analog		-0.0084	0	0	+8.8*	+14.3	0	0	0	-2.7
EIRC_44_SP_SlewRateAbs	EIRC 44 SP SlewRateAbs		0	0	0	+0.06*	-0.001	0	0	0	-0.00042
EIRC_44_Setp_Min_CellVoltage	EIRC 44 Setp Min CellVoltage	V	0	0	0	+0.46*	-0.16	0	0	0	0
EIRC_45_SP_Analog	EIRC 45 SP Analog		-0.0084	0	0	+8.8*	+14.3	0	0	0	-2.7
EIRC_45_SP_SlewRateAbs	EIRC 45 SP SlewRateAbs		0	0	0	+0.06*	0	0	0	0	0
EIRC_45_Setp_Min_CellVoltage	EIRC 45 Setp Min CellVoltage	V	0	0	0	+0.46*	-0.15	0	0	0	0
MV_1_1_2_SetVal_1	MV 1 1 2 SetVal 1	%	0	0	0	0	0	0	0	0	-0.16
MV_1_1_2_SetVal_2	MV 1 1 2 SetVal 2	%	0	0	0	0	0	0	0	0	-0.16
MV_1_1_3_SetVal_1	MV 1 1 3 SetVal 1	%	0	0	0	0	0	0	0	0	-0.18
MV_1_1_3_SetVal_2	MV 1 1 3 SetVal 2	%	0	0	0	0	0	0	0	0	-0.16
MV_1_3_2_SetVal	MV 1 3 2 SetVal	%	+0.002	0	0	-1.0	+0.06	0	-0.0026	0	-0.16
MV_1_4_5_SetVal	MV 1 4 5 SetVal	%	0	0	0	0	0	0	0	0	-0.11
MV_1_5_3_SetVal	MV 1 5 3 SetVal	%	+0.002	0	0	-1.0	+0.06	0	-0.0026	0	-0.16
PLC_hostSetp_ctrl_ctEIRC41_SP_SlewRateAbs	Ramp-rate (slew-rate) setting of the load on string 1		0	0	0	+1.7	-0.06	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC41_SP_TargetValue_Value	Current set-point given to the load on string 1		0	0	0	+3.5	+14.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC41_rCurrMonitor	Current drawn from string 1	A	0	0	0	+0.92	+15.1	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC41_rVoltCellMin_ActV	Lowest cell voltage seen by the load on string 1	V	0	0	+0.00076	+0.61	-0.15	+0.00092	-0.04	-0.0004	+0.05
PLC_hostSetp_ctrl_ctEIRC41_rVoltCellMin_Setp	Setpoint/monitor sent to string 1 load: Setp	V	0	0	0	+0.17	-0.15	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC41_rVoltMonitor	Voltage measured by the electronic load on string 1	V	0	0	0	+863	-96.1	0	0	0	-0.18
PLC_hostSetp_ctrl_ctEIRC42_SP_SlewRateAbs	Ramp-rate (slew-rate) setting of the load on string 2		0	0	0	+1.7	-0.06	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC42_SP_TargetValue_Value	Current set-point given to the load on string 2		0	0	0	+3.5	+14.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC42_rCurrMonitor	Current drawn from string 2	A	0	0	0	+0.70	+15.4	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC42_rVoltCellMin_ActV	Lowest cell voltage seen by the load on string 2	V	0	0	+0.00079	+0.61	-0.16	+0.003	-0.04	-0.0004	+0.05

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
PLC_hostSetp_ctrl_ctEIRC42_rVoltCellMin_Setp	Setpoint/monitor sent to string 2 load: Setp	V	0	0	0	+0.17	-0.15	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC42_rVoltMonitor	Voltage measured by the electronic load on string 2	V	-0.0012	+5.9e-05	-0.0005	+868	-96.9	-0.0011	+0.0011	-0.00039	-0.87
PLC_hostSetp_ctrl_ctEIRC43_SP_TargetValue_Value	Current set-point given to the load on string 3		0	0	0	+1.3	+14.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC43_rCurrMonitor	Current drawn from string 3	A	0	0	0	+1.2	+15.5	0	+1.1e-07	+6.6e-08	0
PLC_hostSetp_ctrl_ctEIRC43_rVoltCellMin_ActV	Lowest cell voltage seen by the load on string 3	V	0	0	+0.00057	+0.49	-0.12	0	-0.05	-0.0004	+0.05
PLC_hostSetp_ctrl_ctEIRC43_rVoltCellMin_Setp	Setpoint/monitor sent to string 3 load: Setp	V	0	0	0	+0.17	-0.15	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC43_rVoltMonitor	Voltage measured by the electronic load on string 3	V	0	0	0	+864	-91.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC44_SP_SlewRateAbs	Ramp-rate (slew-rate) setting of the load on string 4		0	0	0	+1.7	-0.06	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC44_SP_TargetValue_Value	Current set-point given to the load on string 4		0	0	0	+3.5	+14.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC44_rCurrMonitor	Current drawn from string 4	A	0	0	0	+0.74	+11.5	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC44_rVoltCellMin_ActV	Lowest cell voltage seen by the load on string 4	V	0	0	+0.00042	+0.59	-0.17	+0.003	-0.04	-4.4e-05	+0.05
PLC_hostSetp_ctrl_ctEIRC44_rVoltCellMin_Setp	Setpoint/monitor sent to string 4 load: Setp	V	0	0	0	+0.17	-0.16	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC44_rVoltMonitor	Voltage measured by the electronic load on string 4	V	0	0	0	+864	-94.1	0	0	0	-0.30
PLC_hostSetp_ctrl_ctEIRC45_SP_SlewRateAbs	Ramp-rate (slew-rate) setting of the load on string 5		0	0	0	+1.7	0	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC45_SP_TargetValue_Value	Current set-point given to the load on string 5		0	0	0	+3.5	+14.3	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC45_rCurrMonitor	Current drawn from string 5	A	0	0	0	+0.71	+14.8	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC45_rVoltCellMin_Setp	Setpoint/monitor sent to string 5 load: Setp	V	0	0	0	+0.17	-0.15	0	0	0	0
PLC_hostSetp_ctrl_ctEIRC45_rVoltMonitor	Voltage measured by the electronic load on string 5	V	0	0	0	+884	-97.4	0	0	0	-0.10
ct_FIRC_113_OUT_CurrentPID_SetPoint	Controller output of FIRC 113	A	0	0	0	0*	+509	0	0	0	-0.78
ct_FIRC_113_SP_Analog	ct FIRC 113 SP Analog	ln/min	0	0	0	0*	+509	0	0	0	-0.78
ct_FIRC_12_SP_Analog	ct FIRC 12 SP Analog	ln/min	0	0	0	+348	-194	0	0	0	0
ct_TIRC_517_SP_TargetValue	Temperature controller 517 measured value	°C	0	0	+0.22	0	-103	0	0	0	0
hostSetp_ctAFSR_ActFuelSupply	hostSetp ctAFSR ActFuelSupply		0	0	0	+94.2	+41.2	0	0	0	0
hostSetp_ctAFSR_SO_OutValue	hostSetp ctAFSR SO OutValue		0	0	+0.0005	-1.6	-2.5	0	0	0	0
pi_EIRC_2_01_ActV_rMas_EffSetpoint_[%]	Heater power controller (EAPH): ActV rMas EffSetpoint [%]	%	+9.9*	-2.1	+1.3	-47.5*	+25.4	0*	0	0*	-21.6
Insulation monitors											
B0_0x10F1_DC_Offset/[%]	Insulation monitor B0: DC_Offset [%]	%	-0.34	-0.36	+0.0048	-0.90	-2.9	+0.42	+0.24	+0.27	+0.19
B0_0x2008(ohTestStat)_Uac_L1L3/[V]	Insulation monitor B0: Uac_L1L3 [V]	V	+1.6	+13.2	+0.22	+197	-59.3	+10.3	-9.8	-14.1	-8.1
B0_0x200C(ohTestStat)_Uac_L1PE/[V]	Insulation monitor B0: Uac_L1PE [V]	V	+0.36	+4.6	+0.14	+92.6	-50.1	+7.7	-4.2	-7.2	-28.3
B0_0x2010(ohTestStat)_Uac_L3PE/[V]	Insulation monitor B0: Uac_L3PE [V]	V	+0.23	+9.3	+0.16	+114	-36.8	+0.33	-6.2	-7.2	-42.1
B0_0x2026(ohTest)_DurationMeasImpuls/[%]	Insulation monitor B0: B0_0x2026(ohTest)_DurationMeasImpuls [%]	%	+0.10	+0.80	-0.01	-31.3	+25.0	-0.20	-0.49	-0.01	-55.4
B0_0x2028(ohTestStat)_Udc_To_PE/[V]	Insulation monitor B0: Udc_To_PE [V]	V	0	-3.3	-0.05	-14.1	-9.4	+3.3	+1.3	0	+5.7
B1_0x10F1_DC_Offset/[%]	Insulation monitor B1: DC_Offset [%]	%	-0.27	-0.29	+0.03	+0.17	-6.5	+0.64	+0.14	+0.30	-1.4
B1_0x2008(ohTestStat)_Uac_L1L3/[V]	Insulation monitor B1: Uac_L1L3 [V]	V	+1.9	+11.1	+0.15	+173	-55.8	+10.4	-10.1	-12.8	-15.4
B1_0x200C(ohTestStat)_Uac_L1PE/[V]	Insulation monitor B1: Uac_L1PE [V]	V	+0.48	+5.5	+0.24	+83.5	-68.6	+8.2	-2.3	-6.6	-51.8
B1_0x2010(ohTestStat)_Uac_L3PE/[V]	Insulation monitor B1: Uac_L3PE [V]	V	+0.35	+7.6	-0.09	+92.4	-19.3	+0.66	-5.4	-6.4	-42.1

Curve	What it is	Unit	Heat-up air	Heat-up N2+H2	Hot standby	Fuel switch	Part load	Cooling air 6000	Cooling air 8-10k	Shutdown (hot)	Trip purge
B1_0x2026(ohTest)_DurationMeasImpuls/[%]	Insulation monitor B1: B1_0x2026(ohTest)_DurationMeasImpuls [%]	%	-0.13	+1.5	-0.06	+1.0	-2.2	+0.77	-0.66	+0.02	+4.4
B1_0x2028(ohTestStat)_Udc_To_PE/[V]	Insulation monitor B1: Udc_To_PE [V]	V	0	-3.4	+0.11	-8.5	-22.5	+2.9	+0.54	0	+1.6
B2_0x10F1_DC_Offset/[%]	Insulation monitor B2: DC_Offset [%]	%	-0.26	-0.60	+0.07	-1.6	-6.1	+1.1	+0.27	+0.32	-3.9
B2_0x1101_MeasQuality/[%]	Insulation monitor B2: MeasQuality [%]	%	0	0	0	0	0	0	0	0	0
B2_0x2008(ohTestStat)_Uac_L1L3/[V]	Insulation monitor B2: Uac_L1L3 [V]	V	+1.7	+9.8	+0.21	+207	-59.1	+9.2	-10.6	-11.9	-6.5
B2_0x200C(ohTestStat)_Uac_L1PE/[V]	Insulation monitor B2: Uac_L1PE [V]	V	+0.46	+2.0	+0.29	+85.0	-69.0	+9.7	-4.7	-6.1	-40.8
B2_0x2010(ohTestStat)_Uac_L3PE/[V]	Insulation monitor B2: Uac_L3PE [V]	V	+0.33	+9.0	+0.16	+118	-22.0	-0.70	-7.6	-6.0	-30.0
B2_0x201E(ohTest)_MeasQuality/[%]	Insulation monitor B2: B2_0x201E(ohTest)_MeasQuality [%]	%	0	0	0	0	0	0	0	0	0
B2_0x2026(ohTest)_DurationMeasImpuls/[%]	Insulation monitor B2: B2_0x2026(ohTest)_DurationMeasImpuls [%]	%	+0.10	+0.56	-0.16	-27.5	+37.0	-0.36	-0.94	+0.18	-100
B2_0x2028(ohTestStat)_Udc_To_PE/[V]	Insulation monitor B2: Udc_To_PE [V]	V	0	-3.6	+0.21	-16.5	-20.6	+8.4	+1.5	0	-13.9
B3_0x10F1_DC_Offset/[%]	Insulation monitor B3: DC_Offset [%]	%	-0.30	-0.51	+0.02	-2.4	-11.6	+1.2	+0.27	+0.32	+2.6
B3_0x2008(ohTestStat)_Uac_L1L3/[V]	Insulation monitor B3: Uac_L1L3 [V]	V	+2.1	+12.8	+0.59	+190	-52.1	+12.0	-10.2	-11.5	-24.2
B3_0x200C(ohTestStat)_Uac_L1PE/[V]	Insulation monitor B3: Uac_L1PE [V]	V	+0.57	+3.1	+0.19	+81.6	-79.1	+10.9	-12.9	-5.9	-19.9
B3_0x2010(ohTestStat)_Uac_L3PE/[V]	Insulation monitor B3: Uac_L3PE [V]	V	+0.45	+9.8	-0.0066	+117	+8.9	+0.10	-15.0	-5.8	-46.0
B3_0x2026(ohTest)_DurationMeasImpuls/[%]	Insulation monitor B3: B3_0x2026(ohTest)_DurationMeasImpuls [%]	%	-0.33	+1.9	+0.05	-14.6	+8.1	+2.8	-0.35	-0.10	+53.8
B3_0x2028(ohTestStat)_Udc_To_PE/[V]	Insulation monitor B3: Udc_To_PE [V]	V	0	-3.9	+0.08	-12.6	-46.2	+7.0	+1.5	0	+12.5
B4_0x10F1_DC_Offset/[%]	Insulation monitor B4: DC_Offset [%]	%	-0.65	-0.42	+0.04	+0.80	-6.5	+1.0	+0.31	+0.36	-3.2
B4_0x2008(ohTestStat)_Uac_L1L3/[V]	Insulation monitor B4: Uac_L1L3 [V]	V	+3.1	+9.5	+0.19	+111	-51.9	+6.9	-9.5	-12.7	-9.8
B4_0x200C(ohTestStat)_Uac_L1PE/[V]	Insulation monitor B4: Uac_L1PE [V]	V	+0.80	+2.2	+0.26	+38.6	-59.9	+7.8	-12.9	-6.5	-17.1
B4_0x2010(ohTestStat)_Uac_L3PE/[V]	Insulation monitor B4: Uac_L3PE [V]	V	+0.65	+8.3	+0.06	+44.1	-16.2	-0.39	-15.7	-6.4	+24.1
B4_0x2026(ohTest)_DurationMeasImpuls/[%]	Insulation monitor B4: B4_0x2026(ohTest)_DurationMeasImpuls [%]	%	-0.20	+1.9	+0.05	-4.2	+4.8	+1.4	-0.37	+0.47	-31.5
B4_0x2028(ohTestStat)_Udc_To_PE/[V]	Insulation monitor B4: Udc_To_PE [V]	V	0	-3.4	+0.18	+2.2	-20.7	+6.7	+2.0	0	-16.4

* = only one run. – = no data in that stage.